# D3.1 Reproduction

*Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/ahl-d3-1.html*

*Written for this site from the class slides, with textbook page references for each statement. The chapter is on pages 709–725 of the course textbook. Image credits are on the last slide of the slides for this subtopic.*

## GAMETES

### D3.1.13 Puberty

pp. 709–710

In late childhood the hypothalamus releases more **GnRH**, so the pituitary releases more LH and FSH. These make the testes secrete testosterone and the ovaries secrete oestradiol and progesterone, which bring about the changes of puberty.

### D3.1.14 Spermatogenesis and oogenesis

pp. 710–715

**Spermatogenesis and oogenesis compared** D3.1.14

| | Spermatogenesis | Oogenesis |
|---|---|---|
| Where | testes | ovaries |
| Gametes from one cell | 4 | 1 (plus polar bodies) |
| Division of cytoplasm | equal | unequal |
| When | puberty onwards, continuous | starts before birth; one a month until menopause |
| Number made | millions per day | about 400 in a lifetime |

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Both have mitosis, growth, two divisions of meiosis and differentiation. Each primary spermatocyte gives four small sperm, millions a day for life. Each primary oocyte gives one large egg, because the cytoplasm divides unequally and polar bodies break down; meiosis pauses twice and finishes only at fertilization.

### D3.1.15 Preventing polyspermy

pp. 716–717

In the **acrosome reaction**, enzymes from the sperm head digest a path through the zona pellucida. Once one sperm fuses with the egg, the **cortical reaction** releases enzymes that harden the zona pellucida, so no other sperm can enter.

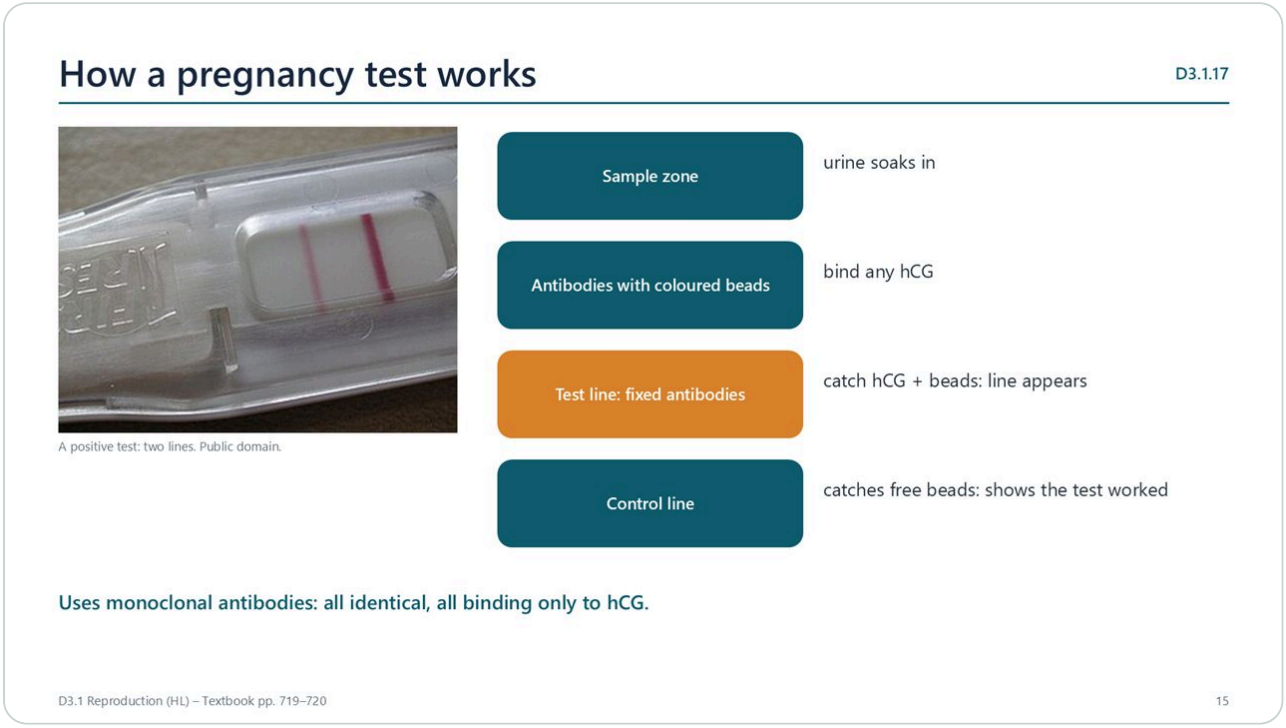
### D3.1.16 Blastocyst and implantation

p. 718

The zygote divides by mitosis as it travels down the oviduct and becomes a hollow **blastocyst**. About a week after fertilization it implants in the endometrium; its outer layer helps form the placenta and its inner cell mass becomes the embryo.

### D3.1.17 Pregnancy tests

pp. 719–720




The embryo and young placenta secrete **hCG**, which appears in urine. Monoclonal antibodies on the test strip bind hCG and are trapped at the test line, giving a coloured line; a control line shows the test worked.

### D3.1.18 The placenta

p. 721

Placental villi give a large surface area for exchange: oxygen and nutrients pass to the foetus, carbon dioxide and urea to the mother, without the bloods mixing. The placenta lets the foetus stay in the uterus to a later stage of development.

### D3.1.19 Pregnancy and childbirth

pp. 722–723

**Progesterone** keeps the pregnancy going, first from the corpus luteum (kept alive by hCG) and then from the placenta. Near term progesterone falls, oxytocin is released and causes contractions, which cause more oxytocin release: **positive feedback** until birth.

## EVIDENCE

**D3.1.20** **HRT and heart disease** pp. 724–725

Early observational studies found less coronary heart disease in women taking HRT. Randomized controlled trials later found a small increase. HRT users tended to have higher socioeconomic status, which itself lowers heart disease risk: the correlation was not cause and effect.

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